

Evaporation

Final Lab Report
Unit Operations Lab I
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Group 1

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1. Executive Summary

The objective of this lab is to use a constant 10 wt% concentration solution of water and sugar, consisting of a nonvolatile solute and a volatile solvent, respectively, and performing experiments using that concentration and the makeup water at different pressures to obtain the capacity and economy of the evaporator. The various temperature readings of the system were found using a digital thermometer for three operating pressures. Results were created by plotting economy, capacity, and the heat transfer coefficient against the vacuum pressure. The economy values for each operating pressure of 10, 15, and 20 inches of mercury were 0.344, 0.485, and 0.618 respectively-signifying that there is an increasing positive trend and all economy value are below the value of 1 as needed. Overall, the trends indicate that there is a linear proportional relationship between capacity, economy, and heat transfer coefficient with the vacuum pressures. The capacity, economy, and heat transfer coefficient against the vacuum pressure has a strong positive trend with R^2 values of 0.99, 0.99, 0.87 which indicate that the trials done for each pressure were done with minor disturbances. Moreover, results validate the theory that the increase in pressure will decrease the boiling point of the solution. The steam temperature does not change as much throughout the experiment due to constant pressure which makes the log mean temperature difference increase with vacuum pressure. The results also validate the theory as the LMTD is defined as the driving force for heat transfer. Overall, this experiment was a success and the results matched the corresponding theories presented in this lab as the trends and values found in our results were as expected.

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2. Introduction

Evaporation is conducted by evaporating a portion of the solvent to produce a concentrated solution or thick liquor. [7] Therefore, this is a heat transfer unit operation. The objective of this lab is to use a constant concentration solution of water and sugar, consisting of a nonvolatile solute and a volatile solvent, respectively, and performing experiments using that concentration and the makeup water at different pressures. Some important properties of evaporating liquids are concentration, foaming, temperature sensitivity, scale on heating surfaces, and materials of construction. Evaporation is a very different process in comparison to other similar processes in that the residue is a viscous liquid, the vapor of evaporation does not need to be separated, and emphasis is placed mainly on concentrating a solution. This experiment is significantly important in the engineering industries to determine the capacity and economy on an evaporator using energy balance. Moreover, this helps determine the performance of the evaporator and if there are any energy losses present. There are many industrial applications of evaporators. In the processing of food and dairy products such as tomato purees, milk, and gelatin, evaporators are used to achieve product consistency. Moreover, they are used to concentrate liquid foods to make items such as condensed milk. Similarly, in pharmaceutical companies, the use of evaporators is necessary to remove excess moisture from drugs, thus improving product stability. [8] Also, heat recovery evaporators are used to convert seawater into clean water in desalination plants. [8]

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For this experiment, a variety of three different pressures were used in units of inches of mercury to obtain critical parameters. The water sugar solution was primarily filled in the separator and water was filled in the makeup tank. The flow of cooling water from the water stream valve was opened. After the water was flowing around the system, the pump was turned on by the switch to operate at the various pressures. The stream valve was opened fully, and a

timer of 15 minutes began or until the system reached steady state. The digital thermometer was used to record critical parameters of certain temperature readings in the system along with the volume using the distillate and steam collection flasks. Readings were done with pressures of 10, 15, and 20 inches of mercury.

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3. Theory

Evaporation is the phase change from liquid to vapor. In a vapor, molecules are further separated from one another as compared to a liquid. Since molecules of a substance have some degree of attraction to each other, energy is required to overcome this intermolecular attractive force. This intermolecular attractiveness translates from the molecular level to bulk level as volatility, where a more volatile compound has weaker intermolecular attraction between its molecules and requires less energy to vaporize. Energy is used for the overcoming of the intermolecular attraction between molecules by means of heat and pressure, work, both factors which achieve boiling point and latent heat of vaporization.

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Include the heat balance equations.

Evaporation is used as a separation process that utilizes volatility differences to selectively vaporize a component of a mixture. Generally, a liquid mixture can have different components evaporated out by subjecting it to targeted temperature, the boiling point for that component. The desired product will either remain in the evaporation vessel as liquid or it may leave as vapor and be collected in another vessel.

An evaporator performs a heat transfer operation. The method used in a single evaporator is called single-effect evaporation where the boiling liquid condensates, and the condensate is discarded. It is a simple process but uses steam ineffectively. This lab scale evaporator works with the use of a steam fed heat exchanger and a condenser.

Several equations were used in this experiment. An energy balance was performed on the evaporator to calculate the capacity and economy.

$$q = UA\Delta T \text{ [1]}$$

Where:

q: the rate of heat transfer (J/s)

U : the overall heat transfer coefficient ($\frac{W}{m^2 K}$)

A : the area available for the heat transfer (m^2)

ΔT : the average temperature difference for heat transfer ($^{\circ}C$)

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In this experiment, the economy is the ratio of mass of product over mass of steam used and capacity is the amount of product (condensate) per hour. Therefore, the rate of heat transfer can be written as:

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$$Capacity = \frac{V_d \cdot \rho_w}{t} = \frac{V_d}{t}$$

Where:

V_d : Volume of distillate (mL)

ρ_w : Density of Water (kg/L) = 1 kg/L

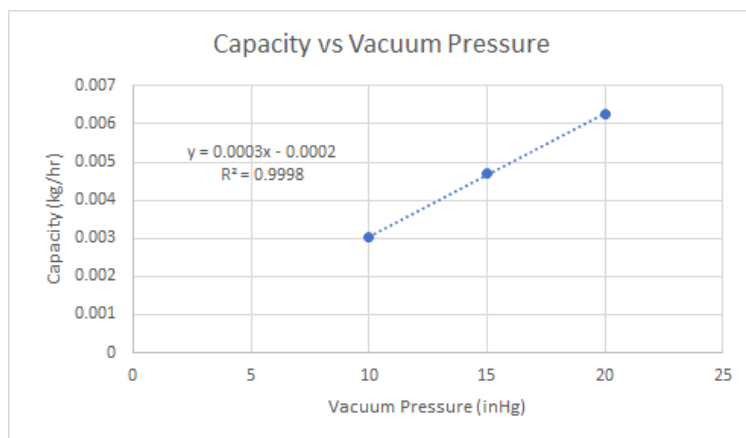
t: Time (hour)

$$Economy = \frac{\text{mass of product (kg)}}{\text{mass of steam used (kg)}}$$

Economy value show the mass of product collected per 1 unit mass of steam used. Therefore, with economy values, estimations and prediction on amount of heat loss in the evaporator or if the feed has entered at its boiling point. However, this value is usually much smaller than 1 for a single effect evaporator. The capacity is about n-times that of a single effect evaporator and the economy is about 0.8n for a n-effect evaporators [6].

4. Results

Economy, capacity and the heat transfer coefficient are plotted against vacuum pressures. This was all done for the 10 wt% sugar water solution.

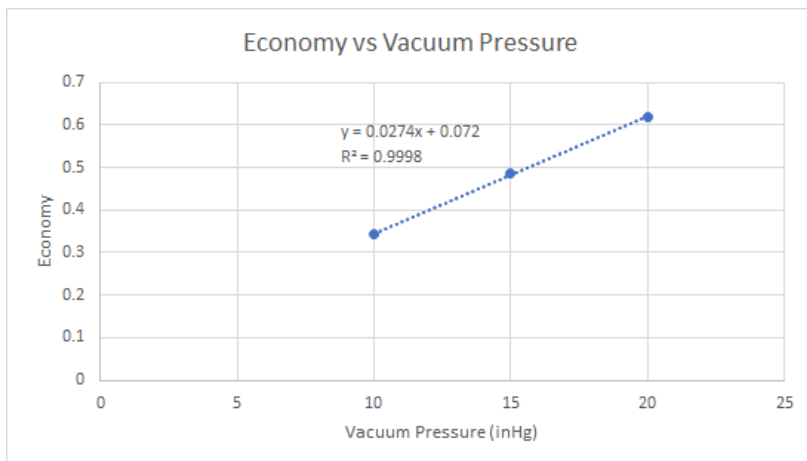


Graph 1: Capacity vs Vacuum Pressure

The Capacity and vacuum pressure have a positive correlation. With an R^2 value of 0.9998 there is a strong linear trend.

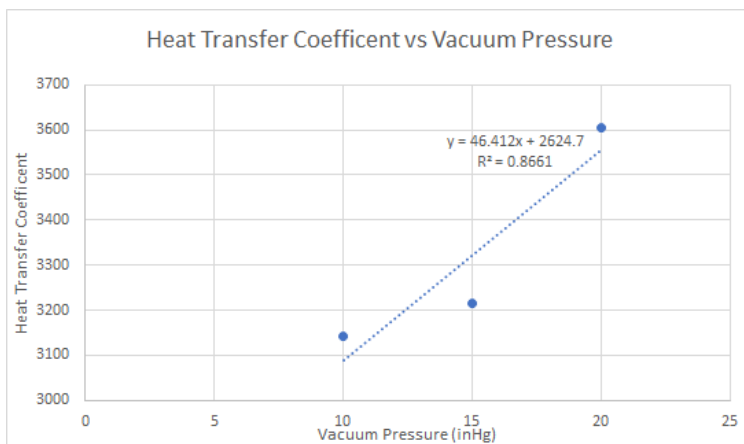
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Include the Vac. Vs. HX Energy and the Vac. Vs. Differential T figures.

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Graph 2: Economy vs Vacuum Pressure

The economy and vacuum pressure also have a positive correlation. The trend is strongly linear with an R^2 of 0.9998.



Graph 3: Heat Transfer Coefficient vs Vacuum Pressure

The economy and vacuum pressure also have a positive correlation. The trend is not as strongly linear with an R^2 of 0.8661.

5. Discussion

Economy is used as a performance basis for industrial evaporators. The economy is defined as the amount of distillate created per unit mass of steam input into the heat exchanger [7]. The economy of a single effect evaporator is always less than 1. In this evaporation experiment, the vacuum pressure put on the liquid in the separator was analyzed. When varying the vacuum pressure, the effects could be seen in the log mean temperature difference (LMTD) between the inlet and outlet heat exchanger, overall heat transfer coefficient, economy, and flowrate from evaporator.

The overall heat transfer coefficient ($W/m^2 \cdot K$) observed was 3141.50, 3215.53 and 3605.62 for respective pressures 10, 15, and 20. The minor variances between the overall heat coefficients can be attributed to random disturbances in the interior of the heat exchanger. These results follow with theory due to the heat transfer coefficient being a function of system sizing, flow conditions, and material [5].

Calculated values of the LMTD were 36.98, 36.13, and 32.22 for vacuum pressure of 10, 15, and 20 respectively. An increase in pressure will decrease the boiling point of a solution, so this increase corresponds with theory. This is due to the fact that vacuum pressure has an opposite effect comparing with positive gauge pressure. When the boiling point decreases with system pressure the temperature must also decrease where the solution undergoes vaporization. The steam temperature at the inlet of the heat exchanger remains the same throughout the experiment due to constant pressure. This results in an increase in LMTD with vacuum pressure. This behavior relates with theory because LMTD is being defined as the driving force for heat transfer [3].

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You should also mention the relationship between the economy and the capacity and the boiling points.

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6. References

- [1] W. McCabe, J. Smith, and P. Harriot, Unit Operations of Chemical Engineering, 7th ed. McGraw-Hill, 2005.
- [2] ChE 381: Unit Operations Laboratory "*Fluid Flow through Fluidized Beds*", rev summer 2014, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.
- [3] Kay J M & Nedderman R M. "*Fluid Mechanics and Transfer Processes*." Cambridge University Press: 1985
- [5] "Raoult's Law; Boiling Point Elevation; Freezing Point Depression." Raoult's Law; Boiling Point Elevation; Freezing Point Depression - CSSAC. N.p., n.d. Web. 3 Feb. 2020.
- [6] NPTEL, NPTEL – Chemical Engineering – Chemical Engineering Design – II Module #3
- [7] Zdunek, Alan. CHE:381 Unit Operations Laboratory. Evaporation, 2014.
- [8] Thermal Kinetics. "Common Uses and Applications for Evaporators." Thermal Kinetics, Thermal Kinetics, 23 Aug. 2019, thermalkinetics.net/common-uses-applications-evaporators.

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[2] and [4] are not found in the report.

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7. Appendix I: Data Tabulation/Graph

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Table 1: Temperature data at different vacuum pressure (°C)

Vacuum (inHg)		10	15	20
Time (hr)		0.259166667	0.25222222	0.25027778
1	HX Top	87	89	91
2	Condenser (Top)	86	89	91
3	Condenser (Bottom)	8	8	12
4	HX (Bottom)	101	101	100
5	Steam Trap (top)	26	27	31
6	steam Trap (Bottom)	6	7	9
7	Steam Trap (Bottom, flask)	33	30	26
8	Inlet Steam Temp	114	114	114

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Table 2: Steam condensate and solvent condensate collected (mL)

Vacuum (inHg)	10	15	20
Steam Condensate (mL)	2320	2475	2570
Solvent Condensate (mL)	800	1200	1590

Table 3: Steam Condensate and solvent

Vacuum (inHg)	10	15	20
Steam Collected per hour (kg/hr)	0.008835	0.009685	0.010135
Solvent collected per hour (kg/hr)	0.003046	0.004695	0.006270

Table 4: Economy, Log Mean, Heat Transfer Coefficient and rate

Vacuum (inHg)	10	15	20
Log Mean Temp Diff	36.98587062	36.1343096	32.2250483
Economy	0.344827586	0.48484848	0.61867704

Heat Transfer Coefficient (W/(m ² *K))	3141.504799	3215.53923	3605.62035
Heat Transfer Rate (kJ/hr)	243.5303684	266.95382	279.354143

8. Appendix II: Error Analysis

In perspective to the experiment of liquid evaporation human error and equipment error are considered as issues that affect these results. In this experiment, human error in measurement of liquid data gathering could yield due to perspective of the eye. Accounted from the data gathering of the steam condensate and solvent volumes; these data points are measured in graduated cylinders. The graduated cylinders were used for fluid volume measurements many times for the evaporation experiment. While creating the sugar and water mixture, having to measure an exact two liters of water with a one-thousand milliliter graduated cylinder. This yields human error in which reading the graduated cylinder meniscus should not be read nor collected, but eye level of the dipping should be read and collected for dilutions and condensate and solvent volumes should be read. The bottom of the meniscus is collected for data results.

Valves and adjustments by human error are error analysis. Timing of adjusting these valves are important to experiment quantitative data collection. Either with proper or ill proper adjusting to the process would contribute to later calculation. Cooling water valve and pressure valves could shift to lower state valve placements that would affect experiment thus, lab testers should always keep tentative to these issues. Simultaneously, the condensed vapor steam and condensed steam flow valves are opened at the same time; this could show inaccuracy if not done properly.

Equipment error analysis is considered in the vibration of the pump and how it affects the stability of valves. Another error could be based on the thermometer that was used during this experiment, the channels that were measured, they were assumed to be at a system of steady state. True temperatures could be 89.4C, when the readings could just show 89C. The accuracy of thermometer temperature reading was presumed to be whole numbers instead of holding decimal points, creating issues in later calculations. Also, the thermocouple is not hundred percent proof accurate since readings of temperature during fluid flow and vapor flow through heat exchange is fluctuation in the system.

The tanks used in this testing of evaporation experiment was to offset evaporated water in evaporation tank. A constant addition of water is need for the tank. Without constant water, concertation of sugar could be assumed in the system. Without water addition it could ruin the experiment.

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Quantitative analysis is needed. What if the solution level varies ~5%? Try to make error bars according to this.

9. Appendix III: Sample Calculations

Vacuum Pressure: 20

Steam Collected: 2570

Solvent Collected: 1590

Time: 15 mins 1 second

Density: 0.997 g/m³

Unit Conversion:

$$\text{Time : } (15 \text{ min} + 1 \text{ sec}) * \frac{1 \text{ min}}{60 \text{ sec}} * \frac{1 \text{ hour}}{60 \text{ min}} = 0.2502 \text{ hr}$$

Mass flowrate for Steam and Distillate Calculation:

$$\text{steam: } \left(2570 \text{ mL} * \frac{1 \text{ L}}{1000 \text{ mL}} \right) * 0.997 \frac{\text{g}}{\text{m}^3} * \frac{1 \text{ kg}}{1000 \text{ g}} * \frac{1}{0.25 \text{ hr}} = 0.01023 \text{ kg/hr}$$

$$\text{solvent: } \left(1590 \text{ mL} * \frac{1 \text{ L}}{1000 \text{ mL}} \right) * 0.997 \frac{\text{g}}{\text{m}^3} * \frac{1 \text{ kg}}{1000 \text{ g}} * \frac{1}{0.25 \text{ hr}} = 0.00633 \text{ kg/hr}$$

Economy Calculation:

$$\text{Economy: } \frac{\text{solvent collected } \frac{\text{kg}}{\text{hr}}}{\text{steam collected } \frac{\text{kg}}{\text{hr}}} = \frac{0.00644}{0.01023} = 0.618$$

Log Mean Temperature Difference:

T1: HX Top = 91

T2: HX Bottom = 100

t1: Condenser Top = 91

t2: Condenser Bottom = 12

$$LMTD = \frac{(T_1 - t_2) - (T_2 - t_1)}{\ln \frac{(T_1 - t_2)}{(T_2 - t_1)}} = (91 - 12) - (100 - 91) / (\ln((91 - 12)/(100 - 91))) = 32.225$$

Heat Transfer Coefficient:

$$U = \frac{Q}{A \Delta T_{LM}} = \frac{(27228.081 \frac{\text{kJ}}{\text{hr}}) (\frac{1 \text{ hr}}{3600 \text{ s}}) (\frac{1000 \text{ J}}{1 \text{ kJ}})}{(0.065 \text{ m}^2) (32.225)} = 3605.62$$

$$\text{Heat Transfer Rate: } q_t = \dot{m}_s C_p \Delta T + \dot{m}_s \Delta H = 0.01023 \frac{\text{kg}}{\text{hr}} * 4.187 \frac{\text{kJ}}{\text{K}} * (112 - 30) + (0.01023 \frac{\text{kg}}{\text{hr}}) * (27228.081 \frac{\text{kJ}}{\text{hr}}) = 282.184 \frac{\text{kJ}}{\text{hr}}$$

10. Appendix IV: Individual Team Contributions

NAME: Yaser Khalid

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	0.5	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0.25	
Procedure (prelab)	0.5	

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Capacity calculation is missed.

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Project Safety Evaluation (prelab)	0.25	
Final Lab Editing	0.5	
Executive Summary (final lab)	0.5	
Introduction (Final lab)	0.5	
Theory (Final Lab)	0	
Results (final lab)	1.5	
Discussion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	0	
Power Point Presentation	0.5	
Total	12	

NAME: Dulce Colindres

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Pre-lab work and experiment execution
Pre-Lab Editing	2	Added content
Experimental Objective (Prelab)	1	Added content
Apparatus (prelab)	0	n/a
Materials & Supplies (prelab)	1	Added content
Procedure (prelab)	0	n/a
Project Safety Evaluation (prelab)	0	n/a
Final Lab Editing	1	
Executive Summary (final lab)	0	n/a
Introduction (Final lab)	0	n/a
Theory (Final Lab)	2	Added content
Results (final lab)	0	n/a
Discussion (final lab)	0	n/a
References (final and/or prelab)	0	n/a
Data Tabulation/Graphs (final)	0	n/a

Error Analysis (final lab)	0	n/a
Sample Calculations (final Lab)	0	n/a
Power Point Presentation	2	Added content
Total	15	

NAME: Advith Vashist

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	0.5	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0.25	
Procedure (prelab)	0.5	
Project Safety Evaluation (prelab)	0.25	
Final Lab Editing	0.5	
Executive Summary (final lab)	0.5	
Introduction (Final lab)	0.5	
Theory (Final Lab)	0	
Results (final lab)	1.5	
Discussion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	0	
Power Point Presentation	0.5	
Total	12	

NAME: Kathy Chavez

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Hands on work
Pre-Lab Editing	1.5	Reviewing parts
Experimental Objective (Prelab)	n/a	n/a

Apparatus (prelab)	1	Labeling Picture of experiment
Materials & Supplies (prelab)	0.5	Reviewing list
Procedure (prelab)	0	n/a
Project Safety Evaluation (prelab)	0.5	Discussion
Final Lab Editing	1	Review grammar
Executive Summary (final lab)	0	n/a
Introduction (Final lab)	0	n/a
Theory (Final Lab)	0	n/a
Results (final lab)	0	n/a
Discussion (final lab)	0	n/a
References (final and/or prelab)	0	n/a
Data Tabulation/Graphs (final)	0	n/a
Error Analysis (final lab)	2	Review of Error, added content
Sample Calculations (final Lab)	0	n/a
Power Point Presentation	0.5	Added content
Total	13.0	

NAME: Marium

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	0.5	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0.25	
Procedure (prelab)	0.5	
Project Safety Evaluation (prelab)	0.25	
Final Lab Editing	0.5	
Executive Summary (final lab)	0.5	
Introduction (Final lab)	0.5	
Theory (Final Lab)	0	
Results (final lab)	1.5	
Discussion (final lab)	0	

References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	0	
Power Point Presentation	0.5	
Total	12	